

Kaustubh Prashant Sadekar

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Education

Ph.D. Computer Science, Portland State University

June 2022 - Present

Department of Computer Science

B.Tech. Electrical Engineering, Pandit Deendayal Petroleum University

August 2014 - May 2018

School Of Technology Department of Electrical Engineering

Skills

Languages and libraries : Python, C++, OpenCV, PyTorch, PyTorch3D, CuPy, libcamera, RawPy, Halide, JAX, Open3D, PCL

Platforms : Blender, Mitsuba3, Meshroom, ROS, Meshlab, Visual Studio, MATLAB, Arduino IDE

Hardware : Intel Realsense D345i, OAK-D, FARO Focus, EinScan Pro, NVIDIA Jetson Nano, ARM-STM32f4, Arduino, Raspberry-Pi

Experience

3D ML Fellow

September 2025 - Ongoing

🏢 HILOS

Manager: Beichen Xie

- Engineered a custom multi-modal 3D generation model leveraging Vision Transformers and CNNs, achieving a 72x speedup (12 min to 10 sec).
- Architected a GenAI pipeline to synthesize a proprietary multi-modal dataset used for 2D/3D semantic segmentation and image-to-image task.
- Designed a semantically controlled Image-to-Image model that eliminated hallucinations and preserved designer intent, delivering a 10x speedup.
- Both models are integrated into the production stack, supporting technical moats around geometrically constrained 2D and 3D generation.

Graduate Research Assistant At Computational Imaging Lab

June 2022 - September 2025

🏢 Portland State University

Advisor: Prof. Atul Ingle

- SPAD ToF ISP: Designed in-pixel histogramming for 32x compression and a robust depth reconstruction algorithm for accurate estimation under high-flux, extreme dead-time distortion.
- SPAD-LiDAR Simulator: Developed a GPU-accelerated simulator achieving a 36x performance speedup.
- Foundation Models: exploring efficient, training-free techniques to leverage foundation models for low-light imaging and compressed SPAD-LiDAR.

Research Fellow At The Computer Vision Imaging and Graphics Lab

August 2020 - March 2022

🏢 Indian Institute of Technology Gandhinagar

PI: Prof. Shanmuganathan Raman

- 3D reconstruction of cultural heritage sites using FARO terrestrial scanner and EinScan structured light scanner. Proposals for research grants
- Point cloud and mesh post-processing pipelines for structured light 3D scans. Custom photogrammetry pipeline for 3D reconstruction

Research Associate At The Autonomous Robots and Multi-robot Systems (ARMS) Lab

May 2019 - August 2020

🏢 Indian Institute of Technology Bombay

PI: Prof. Leena Vachhani

- Streaming and surveillance system for spherical robots using fisheye camera. Created OmniCV - library for omnidirectional cameras 📄 Read Docs

Selected Publications and Patents

High-Flux Count-Free Single-Photon 3D Cameras (**ICCP best paper award**)

ICCP 2026

Kaustubh Sadekar, Vivek K Goyal, David Maier, Atul Ingle | 📄 Project Page

Single-Photon 3D Imaging with Equi-Depth Photon Histograms

ECCV 2024

Kaustubh Sadekar, David Maier, Atul Ingle | 📄 Project Page

3D Sensing with Single-Photon Cameras for Resource-Constrained Applications

CCD Workshop CVPR 2024

Kaustubh Sadekar, David Maier, Atul Ingle | 📄 Poster

Systems and methods for three-dimensional sensing with single photon cameras for resource-constrained applications

pending US Patents

Kaustubh Sadekar, David Maier, Atul Ingle

Methods for scaling a spherical robot (Patent number 507996)

Indian Patents 2024

Leena Vachhani, Vaibhav Nandkumar Kadam, Abhishek Gupta, Kaustubh Sadekar, Animesh Singhal

A robot system with an upwards-facing camera (Patent number 462647)

Indian Patents 2023

Leena Vachhani, Kaustubh Sadekar, Vaibhav Nandkumar Kadam, Animesh Singhal, Abhishek Gupta

Shadow Art Revisited: A Differentiable Rendering Based Approach

WACV 2022

Kaustubh Sadekar, Ashish Tiwari, Shanmuganathan Raman | 📄 Project Page 📄 Paper

Projects

SPCSimLib: Pytorch-based simulator for passive and active single photon cameras

Key-words: Computational imaging, SPADs, single-photon cameras, time-of-flight cameras 📄 Project Page

- Physics-based modeling of SPAD sensor measurements simulating the effect of Poisson randomness, SPAD dead time, and laser characteristics

OpenTorchISP: GPU Accelerated Differentiable Camera ISP Pipeline

Key-words: software-based ISP, rawpy, PyTorch, GPU acceleration 📄 Project Page 📄 Project Report

- Built a vectorized, differentiable PyTorch image signal processor (ISP) pipeline for RAW image processing, delivering ~ 36x speedup.

OmniCV - Library for omnidirectional cameras

Key-words: fisheye lens, wide-lens cameras, camera models, open sourced code 📄 Read Docs 📄 Code

- ROS compatible library with different models of omnidirectional cameras (C++ and Python support). CI-CD workflow using GitHub actions

Affordable Stereo Camera

Key-words: Stereo vision, 3D computer vision, RGBD cameras, low-cost 3D sensing 📄 Project Page

- Stereo camera with USB webcams. Supporting software for stereo calibration, multiple depth estimation algorithms, and RGB-D data processing